Analytical Cumulative Examination
December 2, 2005

1. (60) A solute, S, originally in phase 1, partitions into phase 2. The partition coefficient, K is given as:

   S (in phase 1) <===> S (in phase 2); $K = \frac{[S]_2}{[S]_1}$

   a. If q is defined as the fraction remaining in phase1 after extraction, derive an expression for q relating the fraction remaining in phase 1 to the volumes of the two phases, V_1 and V_2, respectively, and K (hint: if m = total moles of solute, then qm = fraction total moles in phase 1).

   b. What would the relationship look like if 2 extractions with equal volumes, V_2 were performed?

   c. A solute, S has a partition coefficient of 3 between benzene (phase 2) and water (phase 1). Compare the extraction efficiency if 100 mL of an aqueous solution of S is extracted once with 500 mL of benzene, versus the case where the 100 mL aqueous solution of S is extracted 5 times with 100 mL of benzene.

2. (40) Suppose the solute is a weak acid, HA, that ionizes in water to form A^-. Only HA is soluble in phase 2. The distribution coefficient, D is defined as:

   $$D = \frac{[HA]_2}{[HA]_1 + [A^-]_1}$$

   a. Derive an expression for D in terms of K, K_a for HA and the $[H^+]$.

   b. Supposed one needs to separate 2 weak acids, HA_1 and HA_2 with K_a's of 1.8 x 10^{-5} and 5 x 10^{-8}, respectively. Both acids have identical K's for partitioning between an immiscible organic phase and water. Using the expression derived in part a), determine a pH where the D for HA_2 is 10 x that of HA_1.